第三节 多元数量值函数的导数与微分

3.1 偏导数的概念与几何意义

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记为$f_x(x_0,y_0)$, $\left.\dfrac{\partial f}{\partial x}\right|_{(x_0,y_0)}$, $z_x(x_0,y_0)$, 或 $\left.\dfrac{\partial z}{\partial x}\right|_{(x_0,y_0)}$.

类似地, $z = f(x, y)$ 在点$M_0(x_0, y_0)$ 处对y 的偏导数定义为

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- 如果二元函数$z=f(x,y)$在点(x_0,y_0)处对x 与对y 的偏导数都存在, 则称$f(x,y)$ 在点(x_0,y_0)处可偏导.

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因此, 可用一元函数求导数的方法来求二元函数的偏导数, 只不过在求偏导数时需要将另一个变量视为常量.

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注意: $\frac{\partial P}{\partial V}$ 要当做一个整体来对待, 不能像$\frac{\mathrm{d}y}{\mathrm{d}x}$ 看作$\mathrm{d}y$ 与$\mathrm{d}x$ 的微商, ∂P 与∂V 是没有意义的.

例4. 设$f(x,y)=\begin{cases}\dfrac{xy}{x^2+y^2}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$，求$f_x(0,0)$，$f_y(0,0)$.

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此例表明: 函数$f(x,y)$ 在点(x_0,y_0) 处的偏导数存在, 并不能保证函数在该点连续.

例5. 求三元函数$u=\sqrt[z]{\dfrac{y}{x}}$, 求u 的各个偏导数及$\left.\dfrac{\partial u}{\partial x}\right|_{(1,2,1)}$.

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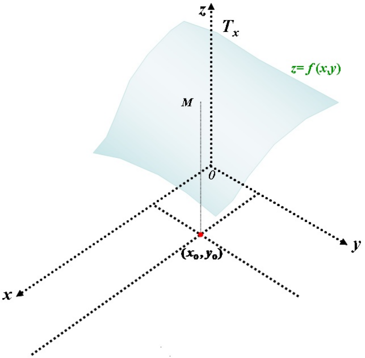

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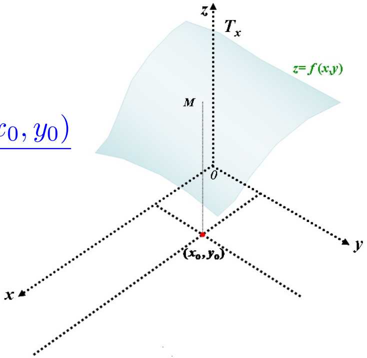

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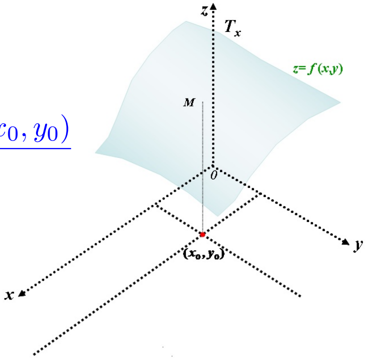

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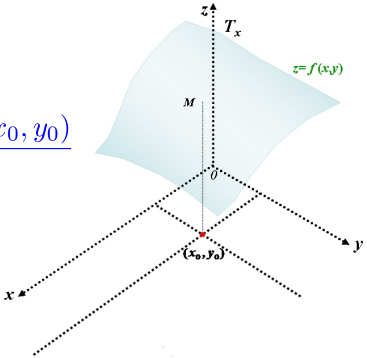

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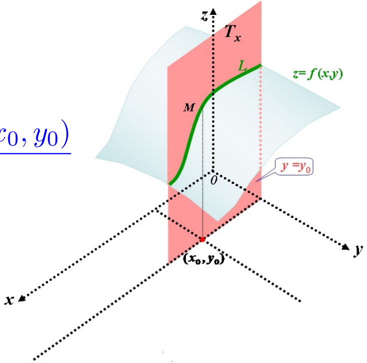

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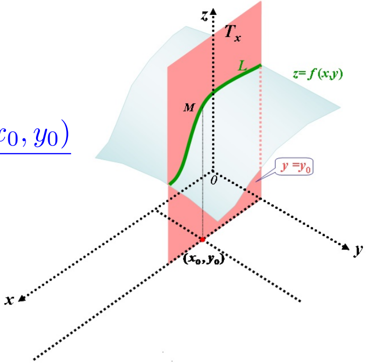

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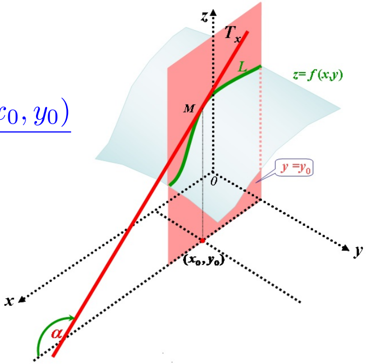

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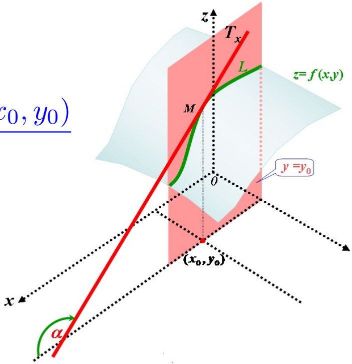

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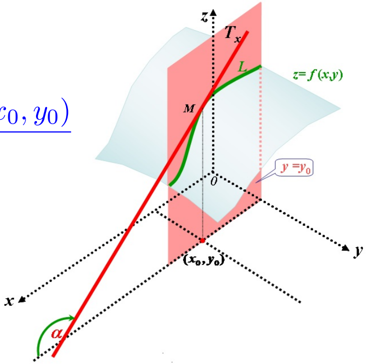

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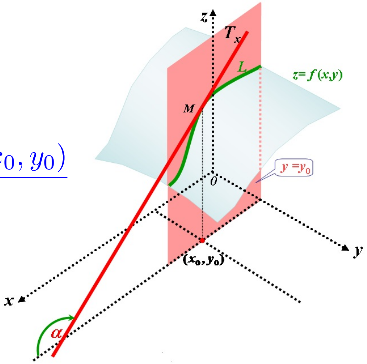

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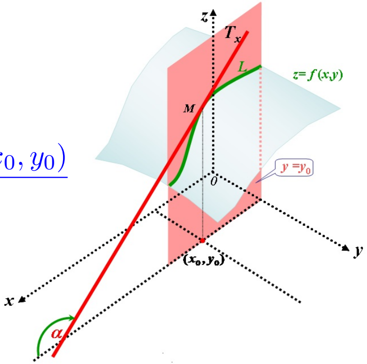

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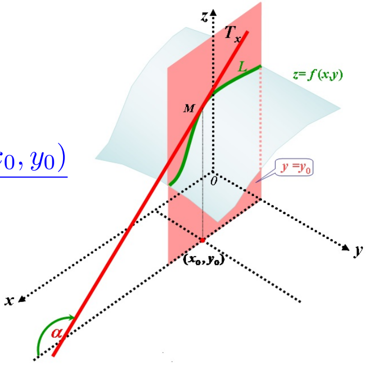

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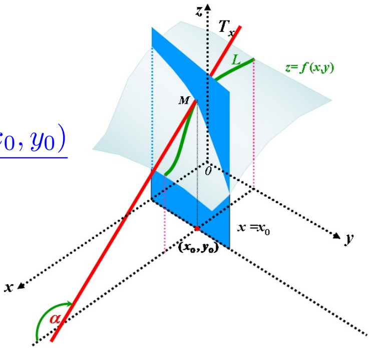

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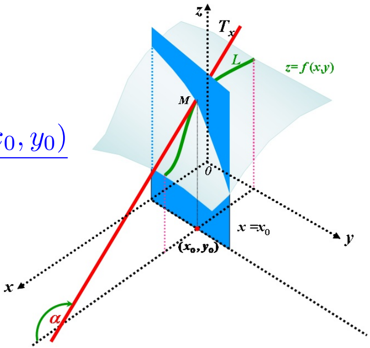

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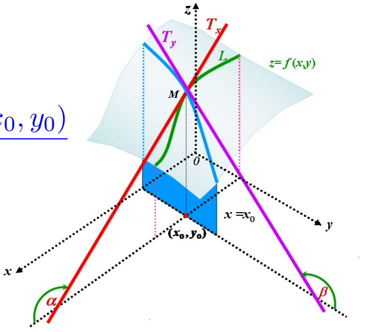

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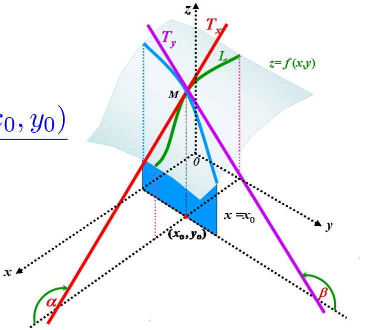

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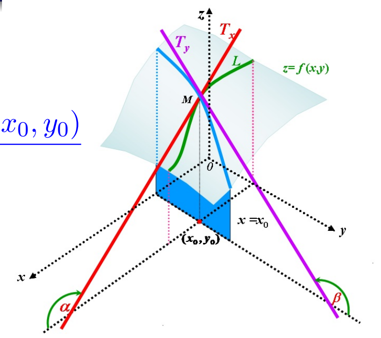

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$$(\frac{\partial z}{\partial y})_x=\frac{\partial}{\partial x}(\frac{\partial z}{\partial y})$$

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设函数$z = f(x, y)$ 在区域D 内具有偏导数$f_x(x, y)$, $f_y(x, y)$, 一般地, 它们仍是x, y 的函数, 若这两个偏导数对x, y 的偏导数也存在, 则称它们为$z = f(x, y)$ 的二阶偏导数.

$$(\frac{\partial z}{\partial x})_y = \frac{\partial}{\partial y}(\frac{\partial z}{\partial x}) = \frac{\partial^2 z}{\partial x \partial y} = f_{xy}(x, y);$$

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其中$f_{xy}(x,y)$ 和$f_{yx}(x,y)$ 称为二阶混合偏导数.

- 同样地, 如果二阶偏导数的偏导数存在, 就称它们为函数 $z = f(x, y)$ 的三阶偏导数.

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- 二阶及二阶以上的偏导数统称为高阶偏导数.

例6. 求$z = x^3y^3 - 3x^2y + xy^2 + 3$ 的二阶偏导数和$\dfrac{\partial^3 z}{\partial x \partial y \partial x}$.

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例7. $f(x,y)=\begin{cases} \dfrac{x^3y}{x^2+y^2}, & x^2+y^2\neq 0 \\ 0, & x^2+y^2=0 \end{cases}$，求 $f_{xy}(0,0)$，$f_{yx}(0,0)$.

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注意: 在例6中, $\dfrac{\partial^2 z}{\partial x \partial y} = \dfrac{\partial^2 z}{\partial y \partial x}$,

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问: 混合偏导数相等需要什么条件?

定理3.1

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如果$f_{xy}(x,y)$, $f_{yx}(x,y)$ 在点(x_0,y_0)某邻域有定义且在(x_0,y_0)处连续, 则有

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注: 此结论可推广到n 元函数高阶导数的情况.

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即: 高阶混合偏导数在连续的条件下与求导次序无关.

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例8. 证明函数$u = \frac{1}{r}$ 满足方程$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ (拉普拉斯方程), 其中$r = \sqrt{x^2 + y^2 + z^2}$.

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证: $\dfrac{\partial u}{\partial x} = -\dfrac{1}{r^2}\dfrac{\partial r}{\partial x} = -\dfrac{1}{r^2} \cdot \dfrac{x}{r} = -\dfrac{x}{r^3}$,

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$$\frac{\partial^2 u}{\partial x^2} = -\frac{1}{r^3} + \frac{3x}{r^4}\frac{\partial r}{\partial x} = -\frac{1}{r^3} + \frac{3x^2}{r^5}.$$

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同理可得 $\dfrac{\partial^2 u}{\partial y^2} = -\dfrac{1}{r^3} + \dfrac{3y^2}{r^5}$, $\quad \dfrac{\partial^2 u}{\partial z^2} = -\dfrac{1}{r^3} + \dfrac{3z^2}{r^5}$,

例8. 证明函数$u=\dfrac{1}{r}$满足方程$\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}+\dfrac{\partial^2 u}{\partial z^2}=0$ (拉普拉斯方程), 其中$r=\sqrt{x^2+y^2+z^2}$.

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$$\therefore\ \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=-\frac{3}{r^3}+\frac{3(x^2+y^2+z^2)}{r^5}$$

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$$\therefore\ \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=-\frac{3}{r^3}+\frac{3(x^2+y^2+z^2)}{r^5}=-\frac{3}{r^3}+\frac{3r^2}{r^5}=0.$$

3.3 全微分

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设函数$z = f(x, y)$ 在点(x_0, y_0) 的某邻域内有定义,

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设函数$z=f(x,y)$ 在点(x_0,y_0) 的某邻域内有定义, $(x_0+\Delta x,y_0+\Delta y)$ 为这邻域内的任意一点, 则称

$$\Delta z=f(x_0+\Delta x,y_0+\Delta y)-f(x_0,y_0)$$

为函数在点(x_0,y_0) 的**全增量**.

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实例:

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实例: 圆柱体体积$V = \pi r^2 h$,

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为函数在点(x_0, y_0) 的**全增量**.

实例: 圆柱体体积$V = \pi r^2 h$, $r \to r + \Delta r$, $\quad h \to h + \Delta h$, 则

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$2\pi rh\Delta r + \pi r^2\Delta h$:

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$2\pi r h \Delta r + \pi r^2 \Delta h$: 关于$\Delta r$ 与Δh 的线性部分.

定义3.2 (全微分)

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设二元函数$z = f(x, y)$ 在点(x_0, y_0) 某邻域$N(x_0, y_0)$ 内有定义.

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则称函数f 在点(x_0,y_0) 处**可微分**, 并称$\alpha\Delta x+\beta\Delta y$ 为函数f

在点(x_0,y_0) 处的**全微分**,

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在点(x_0,y_0) 处的全微分, 记为$\mathrm{d}z\Big|_{(x_0,y_0)}$ 或$\mathrm{d}f(x_0,y_0)$,

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- 由定义可以看出, 当ρ 充分小且α, β 不全为零时, 全微分 $\mathrm{d}z|_{(x_0,y_0)}$ 就是函数f 在(x_0, y_0) 处全增量的线性主部.

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- 若函数$z = f(x, y)$ 在区域D 内处处可微, 则称$z = f(x, y)$ 为区域D 内的可微函数.

定理3.2 (可微的必要条件)

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(2) $f(x,y)$ 在点(x_0,y_0) 处必可偏导, 且$\alpha=f_x(x_0,y_0)$,

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所以$\Delta z=\alpha\Delta x+\beta\Delta y+o(\rho)$, 则$\lim\limits_{\rho\to 0}\Delta z=0$,

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于是

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} f(x_0 + \Delta x, y_0 + \Delta y) = \lim_{\rho\to 0}[f(x_0, y_0) + \Delta z] = f(x_0, y_0),$$

定理3.2 (可微的必要条件)

若$z=f(x,y)$ 在点(x_0,y_0) 处可微, 则

(1) $f(x,y)$ 在点(x_0,y_0) 处连续;

(2) $f(x,y)$ 在点(x_0,y_0) 处必可偏导, 且$\alpha=f_x(x_0,y_0)$,

$\beta=f_y(x_0,y_0)$, 即$\mathrm{d}z=f_x(x_0,y_0)\Delta x+f_y(x_0,y_0)\Delta y$

证明: (1) 因为$z=f(x,y)$ 在点(x_0,y_0) 可微,

所以$\Delta z=\alpha\Delta x+\beta\Delta y+o(\rho)$, 则$\lim\limits_{\rho\to 0}\Delta z=0$,

于是

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} f(x_0+\Delta x,y_0+\Delta y)=\lim_{\rho\to 0}[f(x_0,y_0)+\Delta z]=f(x_0,y_0),$$

$\therefore$ 函数$z=f(x,y)$ 在点(x_0,y_0) 处连续.

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$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = \alpha \Delta x + \beta \Delta y + o(\rho),$$

当$\Delta y = 0$ 时, $\rho = |\Delta x|$, 于是有

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即$f_x(x_0, y_)) = \alpha.$ 同理可得 $f_y(x_0, y_0) = \beta.$

故 $\mathrm{d}z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y.$

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$$\mathrm{d}z=f_x(x_0,y_0)\mathrm{d}x+f_y(x_0,y_0)\mathrm{d}y \quad \text{或} \quad \mathrm{d}z=\frac{\partial z}{\partial x}\mathrm{d}x+\frac{\partial z}{\partial y}\mathrm{d}y.$$

注意: 一元函数中, 可微与可导是等价的. 但在二元函数中, 偏导数存在是可微的必要条件, 而非充分条件,

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$$\text{“}\Delta z-\left(\frac{\partial z}{\partial x}\Delta x+\frac{\partial z}{\partial y}\Delta y\right)\text{是比}\rho\text{高阶的无穷小”}$$

的条件.

例9. 讨论函数$f(x,y)=\begin{cases}\dfrac{x\sin y}{\sqrt{x^2+y^2}}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$ 在点$(0,0)$

处是否可微?

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$$\text{而}\Delta z=f(0+\Delta x,0+\Delta y)-f(0,0)=\frac{\Delta x\cdot\sin(\Delta y)}{\sqrt{(\Delta x)^2+(\Delta y)^2}},$$

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$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \frac{\Delta z-(f_x(x_0,y_0)\Delta x+f_y(x_0,y_0)\Delta y)}{\rho}$$

$$=\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \frac{(f(x_0+\Delta x,y_0+\Delta y)-f(x_0,y_0))-(f_x(x_0,y_0)\Delta x+f_y(x_0,y_0)\Delta y)}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$$

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$$= \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{(f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)) - (f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y)}{\sqrt{(\Delta x)^2 + (\Delta y)^2}}$$

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证明: $\Delta z = f(x_0+\Delta x, y_0+\Delta y) - f(x_0,y_0)$

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$(f(x_0, y_0 + \Delta y) - f(x_0, y_0))$

$= f_x(x_0+\theta_1\Delta x, y_0+\Delta y)\Delta x+f_y(x_0, y_0+\theta_2\Delta y)\Delta y \ (0 < \theta_1, \theta_2 < 1)$

由f_x和f_y在$(x_0, y_))$点连续可得,

$f_x(x_0 + \theta_1\Delta x, y_0 + \Delta y) = f_x(x_0, y_0) + \alpha,$

$$f_y(x_0, y_0 + \theta_2\Delta y) = f_y(x_0, y_0) + \beta,$$

定理3.3 (可微的充分条件)

设函数$z = f(x, y)$ 在点(x_0, y_0) 的某邻域内可偏导, 且$f(x, y)$ 的两个偏导函数$f_x(x, y)$, $f_y(x, y)$ 均在点(x_0, y_0) 连续, 则该函数在点(x_0, y_0) 处可微.

证明: $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$

$= (f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) +$
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$$f_y(x_0, y_0 + \theta_2\Delta y) = f_y(x_0, y_0) + \beta,$$

其中α, β是当$\sqrt{\Delta x^2 + \Delta y^2} \to 0$时的无穷小量.

于是$\Delta z = (f_x(x_0, y_0) + \alpha)\Delta x + (f_y(x_0, y_0) + \beta)\Delta y$

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$$\text{而}0 \leqslant \left|\frac{\alpha\Delta x+\beta\Delta y}{\rho}\right| = \left|\frac{\alpha\Delta x+\beta\Delta y}{\sqrt{(\Delta x)^2+(\Delta y)^2}}\right| \leqslant |\alpha|+|\beta|$$

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所以由全微分定义可知f在点(x_0, y_0)可微.

- 二元函数全微分的定义以及可微分的必要条件和充分条件，可以完全类似地推广到三元和三元以上的多元函数.

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$$\mathrm{d}u=\frac{\partial u}{\partial x}\mathrm{d}x+\frac{\partial u}{\partial y}\mathrm{d}y+\frac{\partial u}{\partial z}\mathrm{d}z.$$

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$$\therefore \mathrm{d}z = 6\mathrm{d}x + 4\mathrm{d}y.$$

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$$\therefore \mathrm{d}u = \mathrm{e}^{x+z}\big((\sin(x+y)+\cos(x+y))\mathrm{d}x + \cos(x+y)\mathrm{d}y + \sin(x+y)\mathrm{d}z\big).$$

例12. 设$f(x,y)=\begin{cases}(x^2+y^2)\sin\dfrac{1}{x^2+y^2}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$，证明:

(1) 在点$(0,0)$ 的邻域内有偏导数$f_x(x,y)$, $f_y(x,y)$;

(2) 偏导数$f_x(x,y)$, $f_y(x,y)$ 在点$(0,0)$ 处不连续;

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证明: (1) 当$x^2+y^2\neq 0$ 时, 有

$$f_x(x,y)=2x\sin\frac{1}{x^2+y^2}-\frac{2x}{x^2+y^2}\cos\frac{1}{x^2+y^2},$$

$$f_y(x,y)=2y\sin\frac{1}{x^2+y^2}-\frac{2y}{x^2+y^2}\cos\frac{1}{x^2+y^2},$$

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故$f(x,y)$ 在点$(0,0)$ 的邻域内有偏导数$f_x(x,y)$，$f_y(x,y)$.

$$(2) \because \lim_{\substack{x\to 0\\ y\to 0}} f_x(x,y) = \lim_{\substack{x\to 0\\ y\to 0}} (2x\sin\frac{1}{x^2+y^2} - \frac{2x}{x^2+y^2}\cos\frac{1}{x^2+y^2})$$

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同理可证, $f_y(x,y)$ 在$(0,0)$ 处不连续.

(3) 因为$\Delta f = f(0+\Delta x, 0+\Delta y) - f(0,0)$

$$= ((\Delta x)^2 + (\Delta y)^2)\sin\frac{1}{(\Delta x)^2 + (\Delta y)^2},$$

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即函数$f(x,y)$ 在点$(0,0)$ 处可微.

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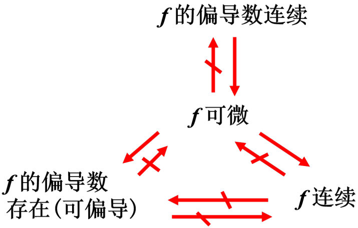

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(2) $\mathrm{d}\left(f(x,y)g(x,y)\right) = g(x,y)\mathrm{d}f(x,y) + f(x,y)\mathrm{d}g(x,y)$;

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与一元函数类似, 多元函数的全微分有以下运算法则:

设 $f(x,y)$, $g(x,y)$ 都是可微函数, 则有

(1) $\mathrm{d}(f(x,y) \pm g(x,y)) = \mathrm{d}f(x,y) \pm \mathrm{d}g(x,y)$;

(2) $\mathrm{d}\left(f(x,y)g(x,y)\right) = g(x,y)\mathrm{d}f(x,y) + f(x,y)\mathrm{d}g(x,y)$;

(3) $\mathrm{d}\dfrac{f(x,y)}{g(x,y)} = \dfrac{g(x,y)\mathrm{d}f(x,y) - f(x,y)\mathrm{d}g(x,y)}{g^2(x,y)} \qquad (g(x,y) \neq 0)$.